NEUROBYTES

BY
TEAM
BIERBYTES

EEG-TO-VIDEO RETRIEVAL

See what the brain watched.

From three seconds of EEG over the visual cortex, NeuroBytes ranks 102 candidate videos and finds the one the person was watching. The model never sees which video was watched: it compares the brain activity with the 102 candidate videos.

36.0%
top-1, chance 0.98%

67.2%
top-5, chance 4.9%

5.99
mean rank of 102

Sushruth Shivaraju and Ayush Shetty
SchubKI National AI Hackathon 2026, EEG mind-reading challenge (Constructor Technology)



THE CHALLENGE

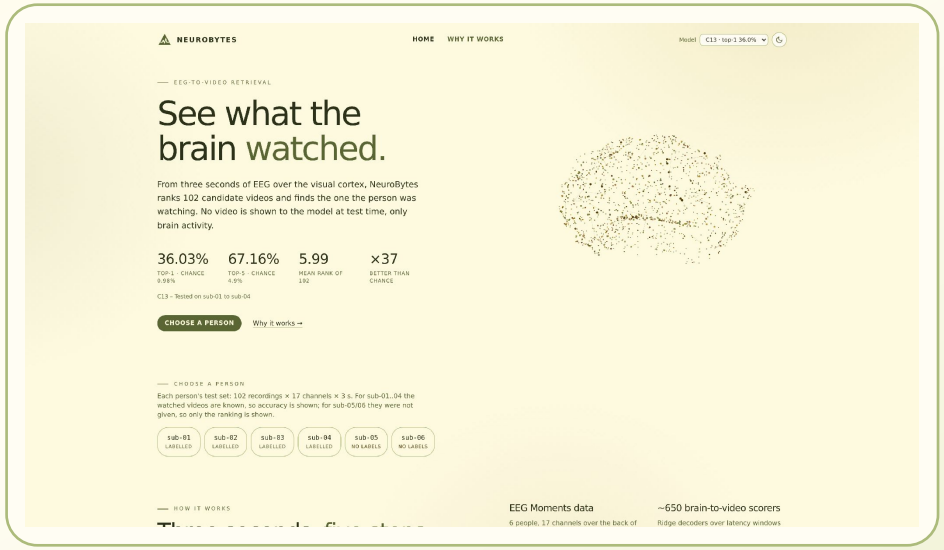
3 seconds of brain activity, 102 possible videos.

3 s
of EEG per recording, while a clip plays

17
posterior channels over the visual cortex

102
candidate videos to rank

408
development recordings, sub-01..04



Chance level
top-1 0.98% | top-5 4.90% | mean rank 51.5 of 102
Data: organizers' EEG Datasets, 6 people, 850-970 training clips each.

HOW IT WORKS

Three seconds, five steps.

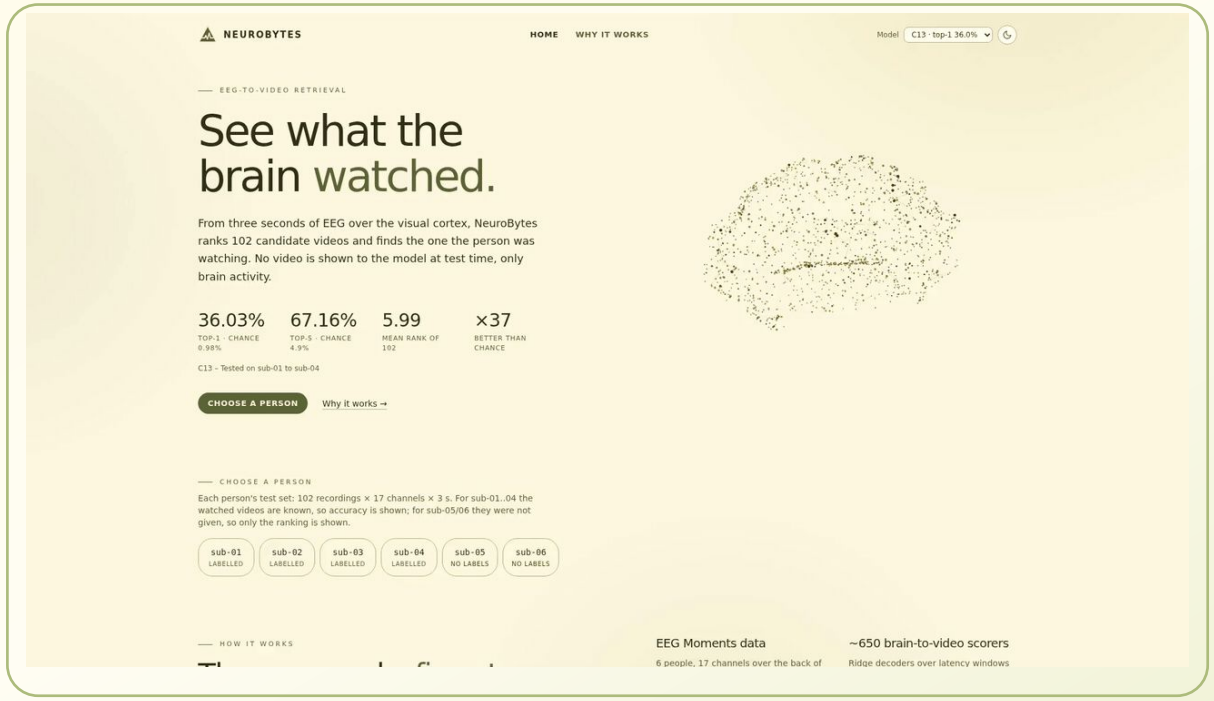
01 Record
3 s of EEG while a clip plays

02 Window
0 to 700 ms after the clip starts

03 Decode
EEG to V-JEPA, CLIP, DINOv2, audio features

04 Fuse
rank fusion of ~590 scorers

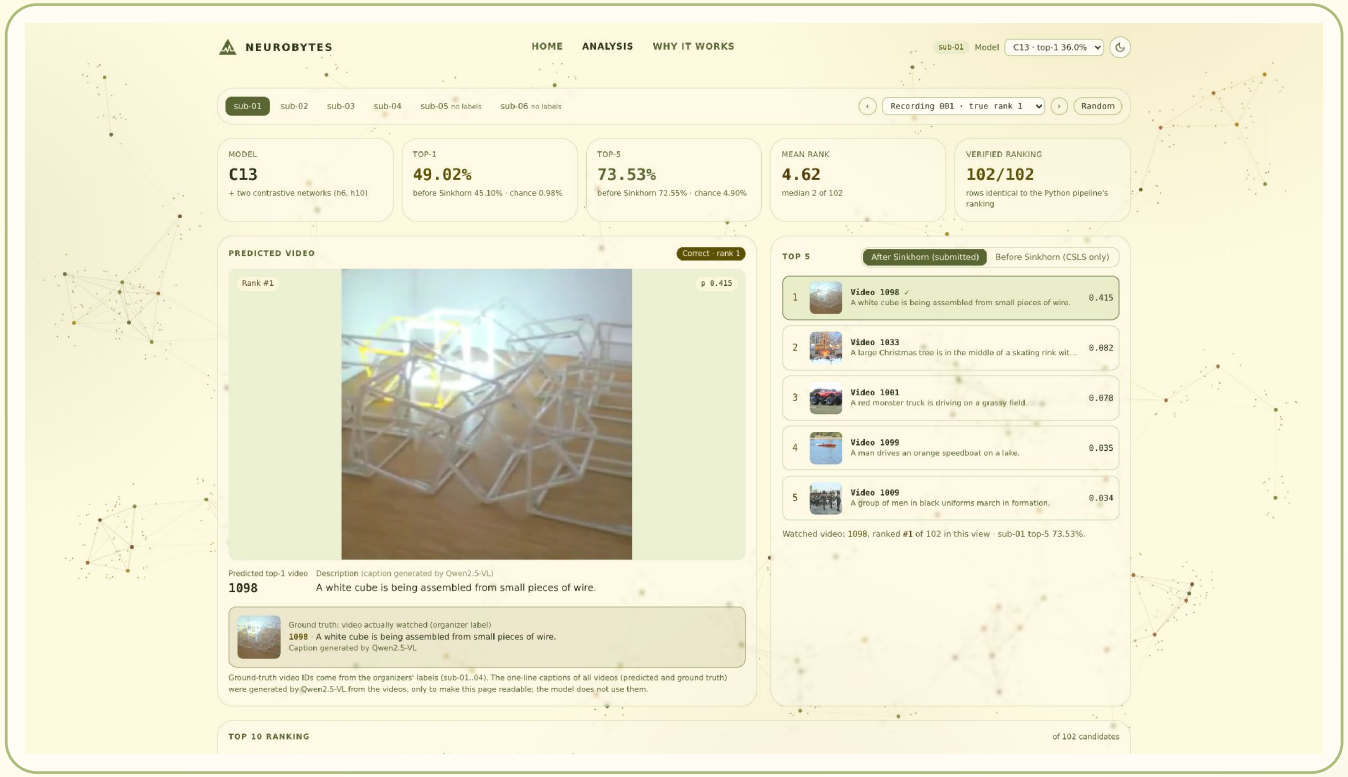
05 Balance
CSLS hub removal + Sinkhorn

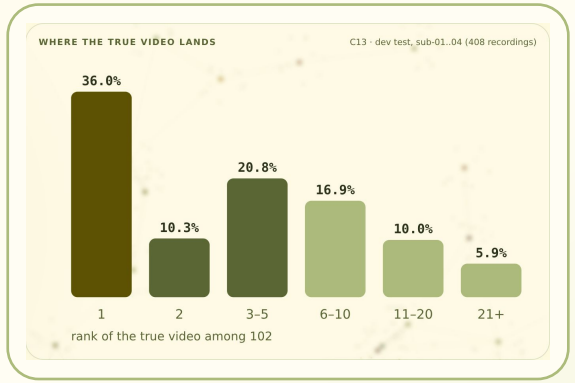


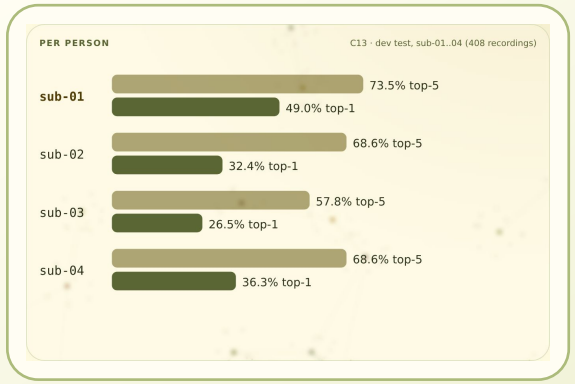
Final Pipeline: C13

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Select a subject and recording to view the score breakdown.







Predicted video with the true clip highlighted. **Top 5** with Qwen2.5-VL captions (display only, not used for ranking). **Per-person ranks** from the same 408 recordings.

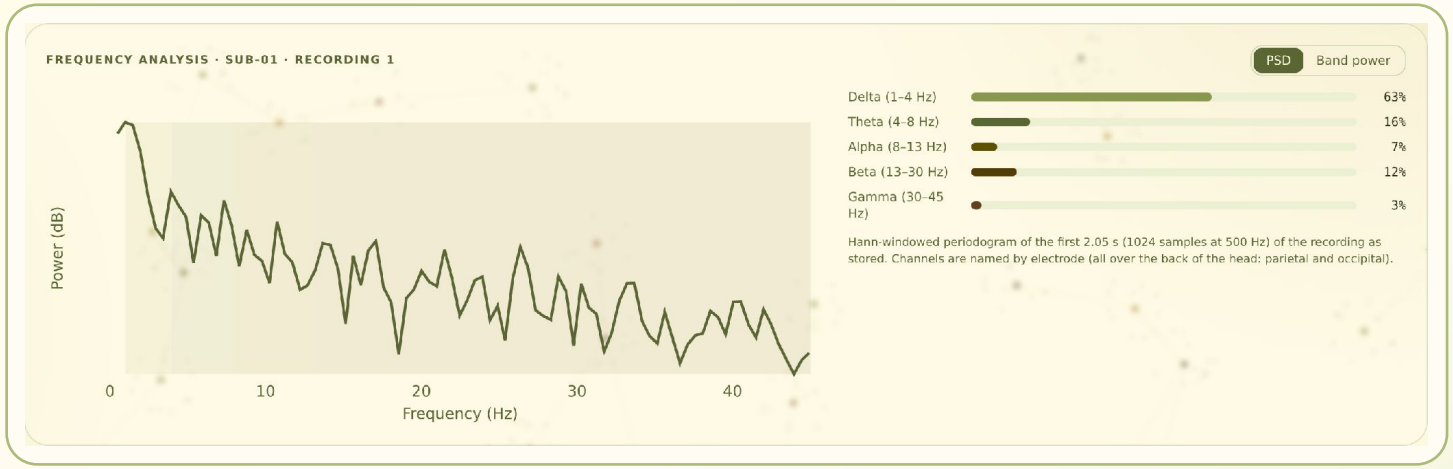
NeuroBytes

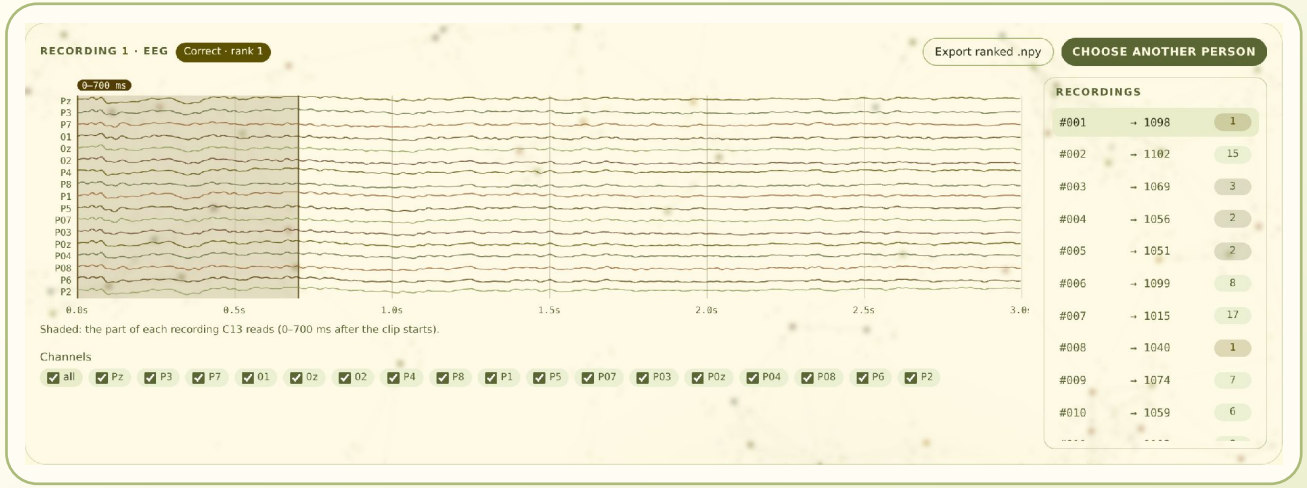
Look at the right moment.

Each recording is 3 seconds long, but almost everything that identifies the video arrives in the first fraction of a second.

The decodable signal sits in the **first 800 ms**, the classic visual ERP window: P1 (about 80 to 130 ms), N1 (about 120 to 200 ms), object categories from 130 to 170 ms, peaking at the N170.

The biggest win came from knowing when to look, not from a bigger video model.





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What holds, and what we disclose.

Run	Top-1 %	Top-5 %	Mean rank
Final C13	**36.03**	**67.16**	**5.99**
Without Sinkhorn	32.35	66.67	6.34
New seeds 5-9	34.31	67.65	5.98
Cross-validation (held-out training videos)	14.95	39.67	16.85
Chance	0.98	4.90	51.5

Same-seed rebuild is bit-identical.

Mean rank 6.00 on the CPU/Docker path. Validation uses held-out training videos and is a harder, different split.

Disclosures

- Development labels (sub-01..04) were used to report results and to compare final candidates.
- EEG z-score statistics were fitted on all of a participant's training recordings, including validation folds (no labels, no test data).
- CSLS and Sinkhorn are batch post-processing over each participant's test recordings; a no-Sinkhorn fallback exists (it still applies CSLS).
- Qwen2.5-VL captions are display only and do not affect the ranking.
- Ranking among 102 known candidates, not unrestricted mind reading.

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Thank you.

Three seconds of EEG, 102 candidates: the right video first 36.0% of the time, in the top 5 67.2% of the time.

Code **github.com/Sushruths04/neurobytes**

Weights **release v1.0**

Docker **ghcr.io/sushruths04/neurobytes**

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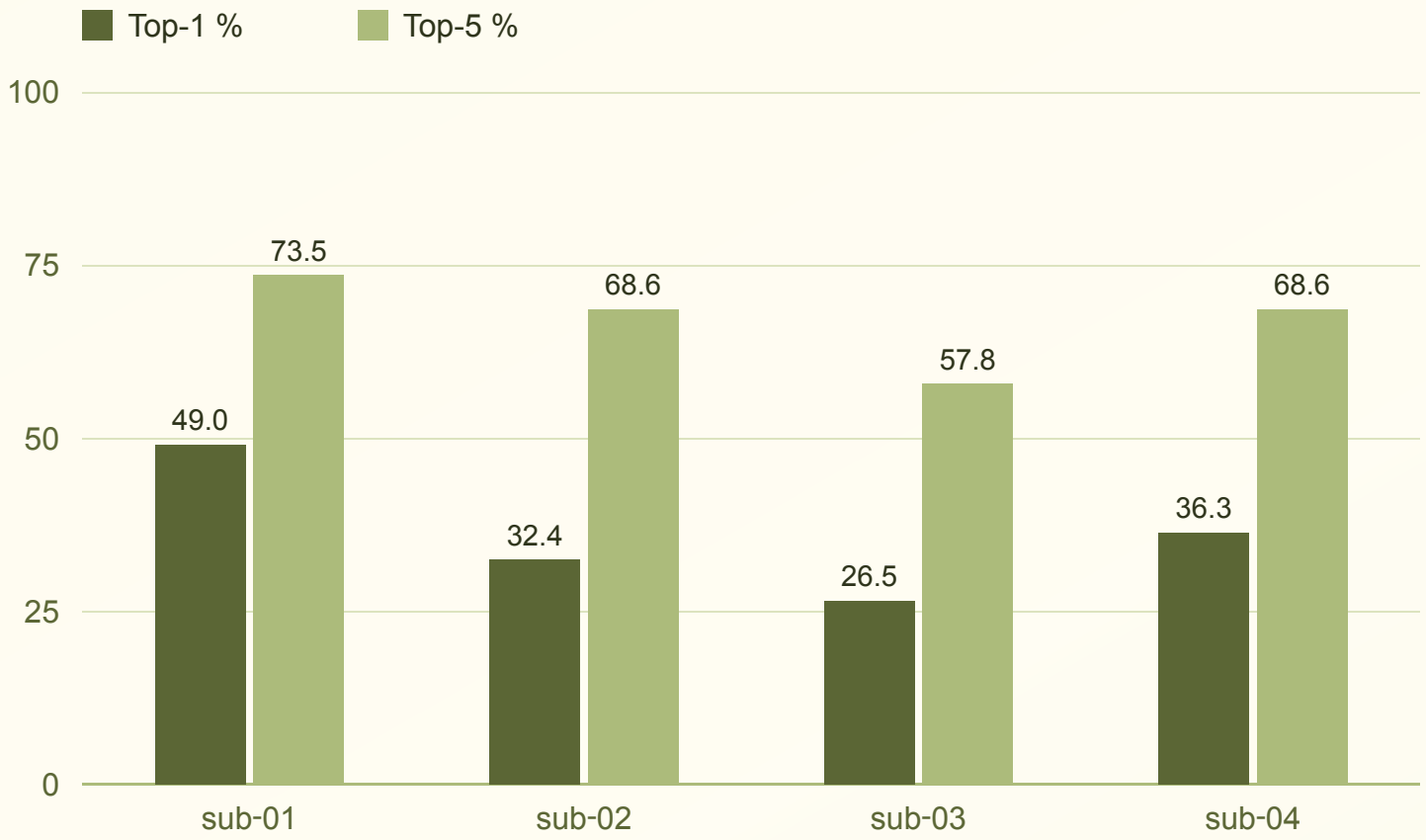


Backup

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RESULTS

Every person well above chance.



36.0%
top-1 pooled, 408 recordings

67.2%
top-5 pooled

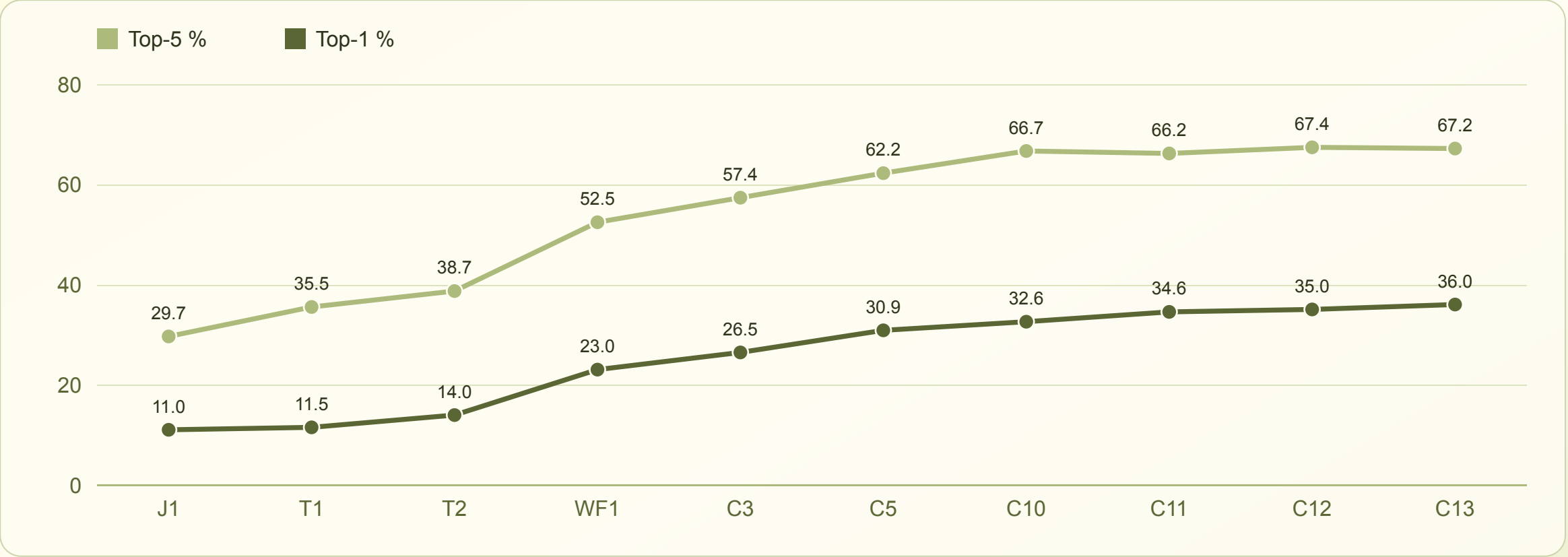
5.99
mean rank of 102 (6.00 CPU/Docker)

Chance: 0.98% top-1, 4.90% top-5, mean rank 51.5. Development test labels were used to report and compare final candidates.

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PROGRESSION

From 11% to 36% top-1, one step at a time.



Development-test top-1 / top-5 (%) per milestone, 408 recordings. J1, T1, T2: single pipelines; WF1: window fusion; C3 to C13: combined candidates. Comparing candidates on development labels is disclosed.